

Surprise Quiz 08

5 question(s) — correct option(s) marked in green.

Q1. What does `sem_wait()` do to a semaphore's value?

- A. Increments it
- B. Decrements it (correct)**
- C. Resets it to 0
- D. Doubles it

Q2. Which semaphore operation is non-blocking?

- A. `sem_wait()`
- B. `sem_post()` (correct)**
- C. Both of the above
- D. None of the above

Q3. A binary semaphore used as a lock should be initialized to what value?

- A. 0
- B. 1 (correct)**
- C. -1
- D. 2

Q4. What is the main problem with using `sleep()` in a loop to wait for a lock to become free?

- A. `sleep()` is not a real syscall
- B. There's no way to know exactly how long to sleep before the lock frees (correct)**
- C. `sleep()` requires root privileges
- D. `sleep()` disables interrupts

Q5. Fix for a race condition on shared data?

- A. Bigger data type
- B. Mutual exclusion (correct)**
- C. Run on separate machines
- D. Mark variable volatile